Fetching Data

PYTHON STRATEGIES FOR WEB SERVICES AND WEB SCRAPING

https://github.com/monicaihli/python-data-workshop

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OBJECTIVES

- Learn to assess use case; Identify & implement appropriate workflow for the situation.
- Learn which libraries + tools will implement the appropriate workflow for a given situation.
- See some simple recipes in action which represent different categories of cases.

Data Over HTTP Workflow

xml.etree.Elementtree
json module
Beautiful Soup 4
lxml

matplotlib
pandas
numpy
seaborn

Reading from files
requests
Selenium + web driver

regex
string functions

Non-Python options:
- prep for database import
- Tableau, Power BI
- R
- Excel

Getting Data

Human Readable

- Information designed for humans, e.g. web pages.
- Visually appealing.
- Less well-structured.
- Typically more work to extract usable data.

Machine Readable

- Exposed through web services / application programming interfaces.
- Machine readable formats (XML, JSON, even CSV/TSV/etc).
- Well-structured, but ugly to look at.

Human Readable

search.dataone.org/view/doi%3A10.18739%2FA2S17ST4H

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Harry McCaughey. 2018. **Carbon dioxide fluxes by the AmeriFlux network at the Ontario - Groundhog River, Boreal Mixedwood Forest site (CA-Gro), 2003-2015.** Arctic Data Center. **doi:10.18739/A2S17ST4H**.

Citations 0 | Downloads 70 | Views 318 | Copy Citation | Assessment report

Files in this dataset Package: resource_map_doi:10.18739/A2S17ST4H

Name		File type	Size		Download All
Metadata: Carbon_dioxide_fluxes_by_the_AmeriFlux_network_at.xml		EML v2.2.0	52 KB		Download
AMF_CA-Gro_BASE_HH_1-1.csv	More info	text/csv	51 MB	8 downloads	Download
AMF_CA-Gro_BIF_LATEST.xlsx	More info	Microsoft Excel OpenXML	25 KB	7 downloads	Download
README_AmeriFlux_BASE.txt	More info	plain text (.txt)	8 KB	55 downloads	Download

General

Machine Readable


```
−<eml:eml packageId="doi:10.18739/A2S17ST4H" xsi:schemaLocation="https://eml.ecoinformatics.org/eml-2.2.0 eml.xsd" scope="system" sy
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    <alternateIdentifier system="DOI">10.17190/AMF/1245996</alternateIdentifier>
   −<title>
      Carbon dioxide fluxes by the AmeriFlux network at the Ontario - Groundhog River, Boreal Mixedwood Forest site (CA-Gro), 2003-2015
    </title>
   −<creator>
    −<individualName>
        <givenName>Harry</givenName>
        <surName>McCaughey</surName>
      </individualName>
      <organizationName>Queen's University</organizationName>
      <electronicMailAddress>mccaughe@queensu.ca</electronicMailAddress>
      <userId directory="https://orcid.org">https://orcid.org/0000-0003-0896-1255</userId>
    </creator>
    <pubDate>2018</pubDate>
   −<abstract>
      Dynamics of carbon dioxide, water, and energy fluxes and the associated meteorological drivers; assessment of the ecological character of t
      sensing to model carbon dioxide exchange and the status of macronutrients in the canopy.
    </abstract>
   −<intellectualRights>
    −<para>
        This work is licensed under the Creative Commons Attribution 4.0 International License.To view a copy of this license, visit http://creative
      </para>
    </intellectualRights>
   −<distribution>
    −<online>
        <url>http://doi.org/doi:10.18739/A2S17ST4H</url>
      </online>
    </distribution>
   −<coverage>
    −<geographicCoverage>
     −<geographicDescription>
        Groundhog River (FCRN or CCP site 'ON-OMW') is situated in a typical boreal mixedwood forest in northeastern Ontario (48.217 degree
```
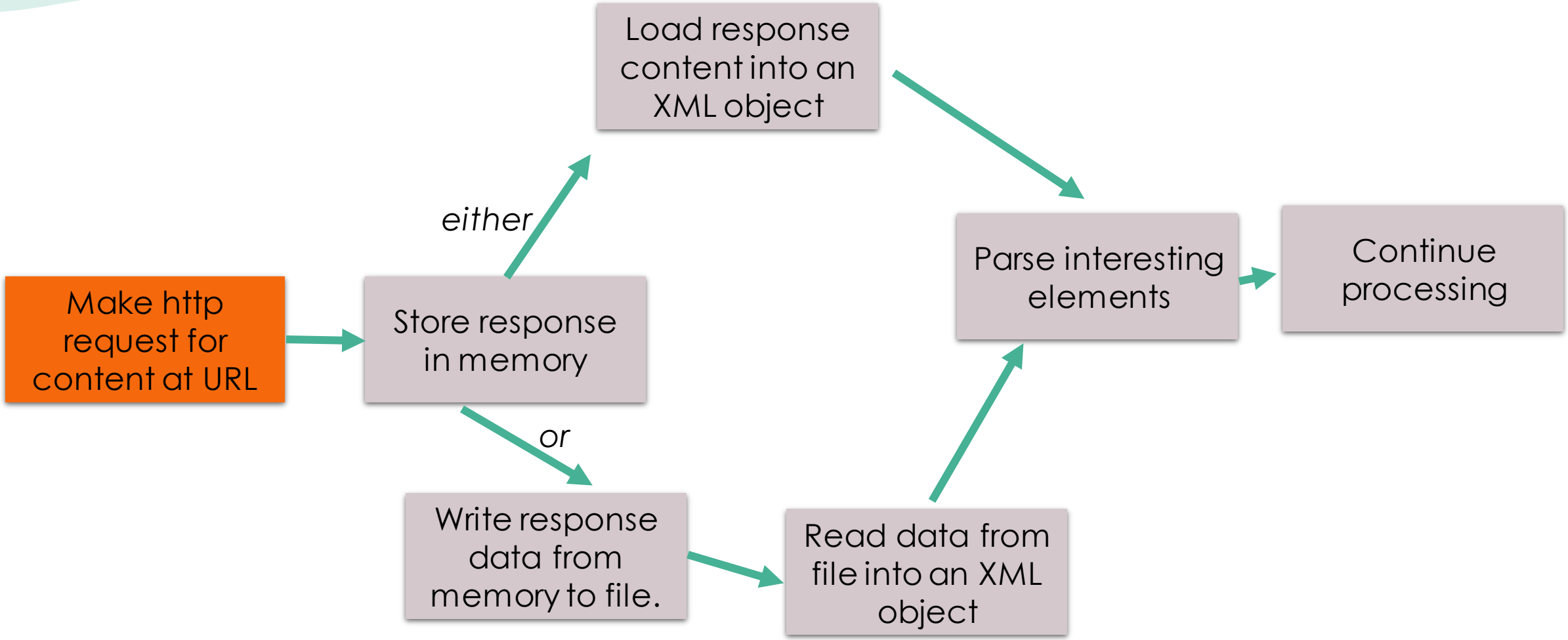
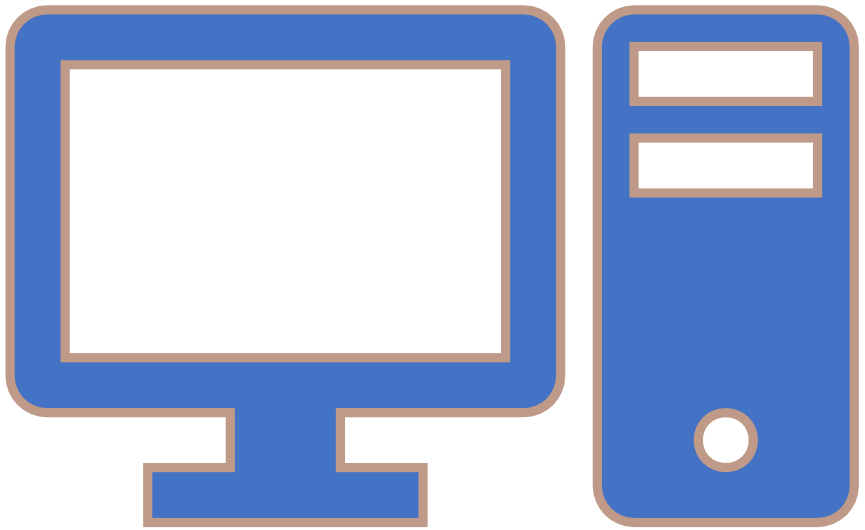

Fetch & Parse XML File on Remote Server



Python requests Library

- Third-party library, must install.
- Convenient third-party library which simplifies handling most aspects of working with **HTTP**.



Some HTTP Basics

HTTP Methods indicate the desired action. Examples:

- **GET** – retrieve the requested resource.
- **POST** - Submit data to the resource.
- **PUT** – Update the target resource with the current payload.

"Payload" refers to the data sent over an HTTP request.

HTTP Headers

GENERAL headers like *Date* or *Connection* apply to request or response, but not to the content of what is being sent.

REQUEST headers like Accept, User-Agent

RESPONSE headers like *Age* or *Server* give additional context about the response.

ENTITY headers like *Content-Length, Content-Encoding* describe content of the message body.

Common Status Codes

200 – Ok *

404 – Not Found

503 – Service unavailable

500 – Internal Server error

* means no error in http request but problems can happen on other levels!

A simple request

```
import requests
r =
requests.get("http://kitty.ninja")
```

```
r = {Response} <Response [200]>
  _content = {bytes} b'Howdy INSC 360 Cadets!\n'
  _content_consumed = {bool} True
  _next = {NoneType} None
  apparent_encoding = {str} 'ascii'
  connection = {HTTPAdapter} <requests.adapters.HTTPAdapter object at 0x7fa06e1b4748>
  content = {bytes} b'Howdy INSC 360 Cadets!\n'
  cookies = {RequestsCookieJar} <RequestsCookieJar[]>
  elapsed = {timedelta} 0:00:00.188703
  encoding = {str} 'ISO-8859-1'
  headers = {CaseInsensitiveDict} {'Date': 'Wed, 25 Mar 2020 20:11:12 GMT', 'Server': 'Apache/2
  history = {list} <class 'list'>: []
  is_permanent_redirect = {bool} False
  is_redirect = {bool} False
  links = {dict} <class 'dict'>: {}
  next = {NoneType} None
  ok = {bool} True
  raw = {HTTPResponse} <urllib3.response.HTTPResponse object at 0x7fa06e1db240>
  reason = {str} 'OK'
  request = {PreparedRequest} <PreparedRequest [GET]>
  status_code = {int} 200
  text = {str} 'Howdy INSC 360 Cadets!\n'
  url = {str} 'http://kitty.ninja/'
```

Request Arguments

- **<u>url</u>**- You can provide a domain-based or even an IP address:

```
url='http://167.99.155.69')
```

- **<u>params</u>** – a dict of key-value pairs representing HTTP queries and inputs:

```
payload = {'verb': 'ListRecords', 'from': '2020-03-25' }
```

Request Parameters

- **headers** – constructed using a dict of key-value pairs:

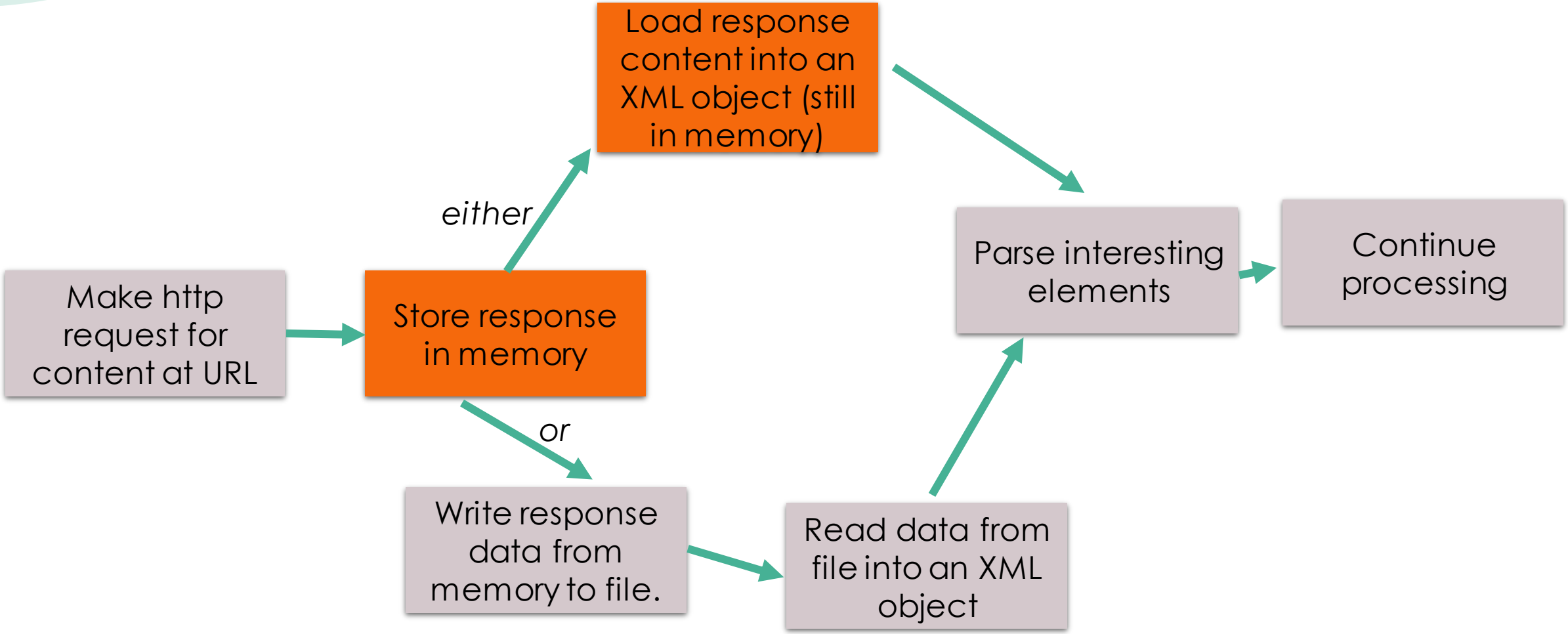
```
headers = {'user-agent': 'Mozilla/5.0 (X11; Ubuntu; Linux x86_64;
rv:74.0) Gecko/20100101 Firefox/74.0'} # haha look like a human!
```

```
r = requests.get(url=baseurl, params=payload, headers=headers)
```

case-1_A.py

- Import requests library.
- Use requests to make an HTTP response to a web server. (http://volweb.utk.edu/~mihli1/demos/page1.html)
- Check response code for expected outcome.
- Access the contents of the server's response (the page you requested).
- View response headers.

Case I:
Fetch & Parse XML File on Remote Server



XML Tools

- XML: eXtensible Markup Language
- Standard Library XML Parser (we use today): *xml.etree.ElementTree*
- Other library options:
  - *lxml* – speedy parsing.
  - *BeautifulSoup* - mainly famous for HTML parsing but also works with an implementation of *lxml.*
  - *xml.dom.minidom* - minimal DOM. Also gives us pretty XML.

Anatomy of a Simple XML Document

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
    <cookies>
        <cookie>chocolate chip</cookie>
        <cookie>oatmeal raisin</cookie>
        <cookie>peanut butter</cookie>
    </cookies>
    <pies>
        <pie>apple</pie>
        <pie>pecan</pie>
        <pie>pumpkin</pie>
    </pies>
</snacks>
```

Document prolog

Anatomy of a Simple XML Document

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
    <cookies>
        <cookie>chocolate chip</cookie>
        <cookie>oatmeal raisin</cookie>
        <cookie>peanut butter</cookie>
    </cookies>
    <pies>
        <pie>apple</pie>
        <pie>pecan</pie>
        <pie>pumpkin</pie>
    </pies>
</snacks>
```

root element

Anatomy of a Simple XML Document

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
  <cookies>
      <cookie>chocolate chip</cookie>
      <cookie>oatmeal raisin</cookie>
      <cookie>peanut butter</cookie>
  </cookies>
  <pies>
      <pie>apple</pie>
      <pie>pecan</pie>
      <pie>pumpkin</pie>
  </pies>
</snacks>
```

example of opening tag

example of closing tag

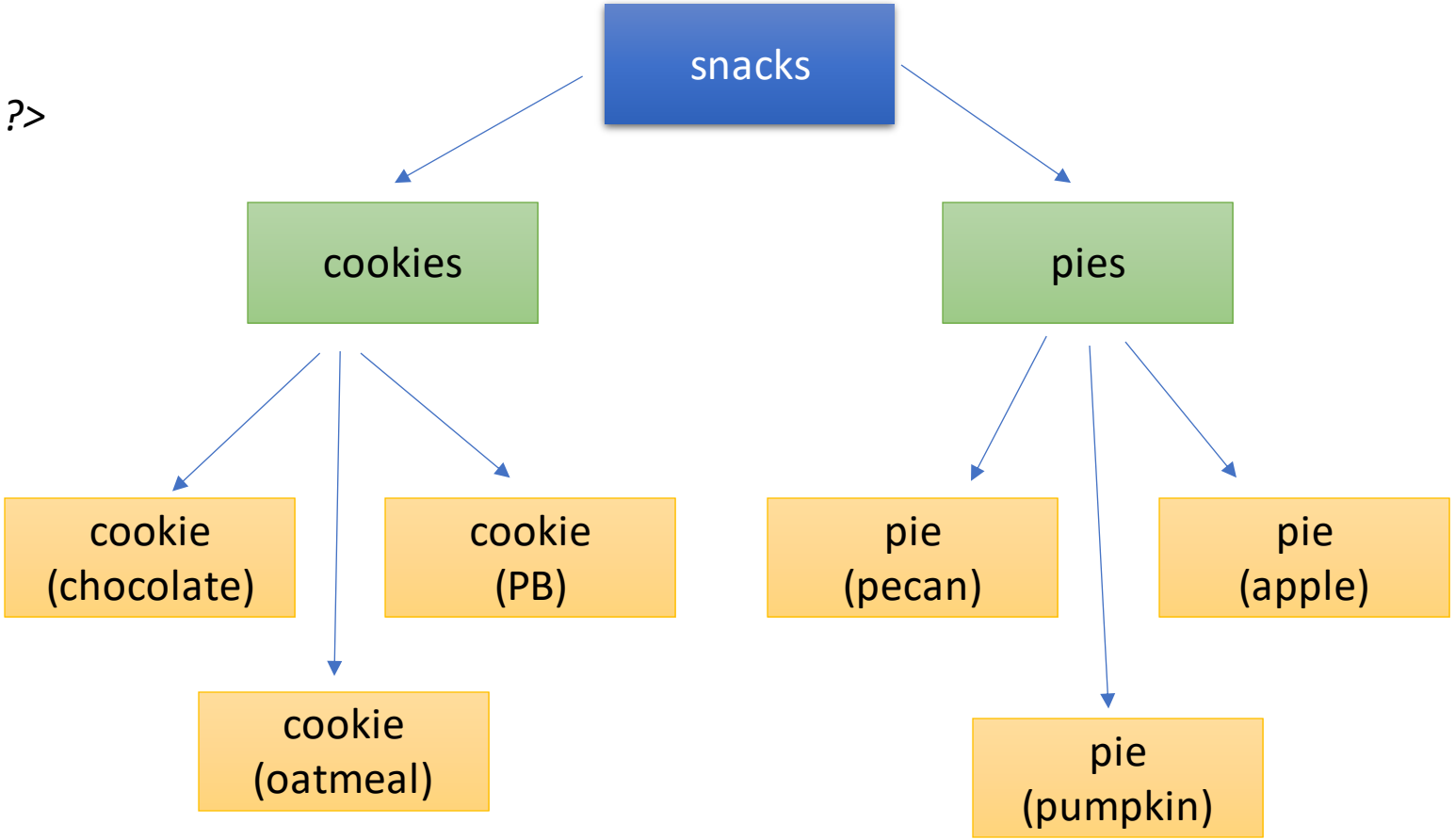
Elements with Attributes

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
    <cookies>
        <cookie sellby="03/14/2020">chocolate chip</cookie>
        <cookie sellby="03/13/2020">oatmeal raisin</cookie>
        <cookie sellby="03/15/2020">peanut butter</cookie>
    </cookies>
    <pies>
        <pie sellby="03/5/2020">apple</pie>
        <pie sellby="03/9/2020">pecan</pie>
        <pie sellby="03/6/2020">pumpkin</pie>
    </pies>
</snacks>
```

- Attributes can be used to store additional information or properties about the element.
- Each attribute can contain one value.

Structure of a Simple XML Document

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
    <cookies>
        <cookie>chocolate chip</cookie>
        <cookie>oatmeal raisin</cookie>
        <cookie>peanut butter</cookie>
    </cookies>
    <pies>
        <pie>apple</pie>
        <pie>pecan</pie>
        <pie>pumpkin</pie>
    </pies>
</snacks>
```



`xml.etree.ElementTree`

- Loads data into a tree-like structure in memory.
- Facilitates both searching for specific elements of data in a document, or iteratively accessing all elements.
- Can modify an existing XML tree , create a new one, and even write XML trees to file.

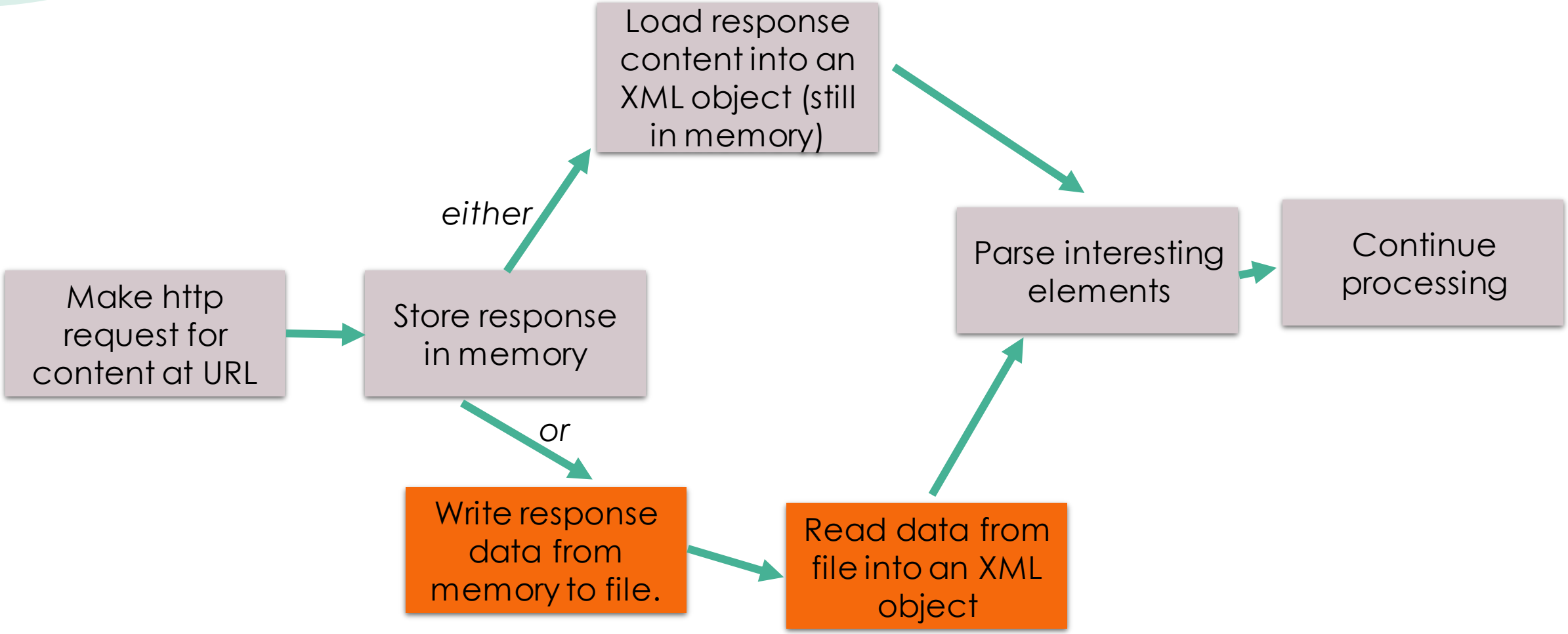
xml.etree.ElementTree

```
# creates a new XML object in memory from data already in memory as string
xml_data = ET.fromstring(r.content)

# creates new XML object in memory by reading from data stored on disk as a file
Xml_data = ET.parse("demo1.xml")
```

case-1_B.py

- Fetch an XML file off a remote server using requests library. (http://volweb.utk.edu/~mihli1/demos/case1.xml)
- Load the contents of the XML document into an element tree object.
- Display the XML elements.

Case 1:
Fetch & Parse XML File on Remote Server



case-1_C.py

- Fetch a set of XML files off a remote server using requests library. (http://volweb.utk.edu/~mihli1/demos/case1-C/file1.xml)
- Write the files to disk
- Cycle through all downloaded XML files in a directory and display them.
- Display the XML elements.

Understanding XML & Parsing w/ Python

`xml.etree.ElementTree`

- Module has two classes for representing the XML data;
  - **ElementTree** – represents the whole document as a tree structure
  - **Element** – represents a single node in the tree (But this node can have children)

- Data can be imported from a string (in memory) or from a file.

- An element can consist of:
  - **tag** – a string identifying the name of the element.
  - **text** – the text contained within the start and end tag, if any.
  - **tail** – in some situations, to capture additional text outside the tag.
  - **attributes** – attributes are stored as a dictionary where the attribute label is the key and the value is the... value.

Strategies for Parsing XML Elements

(1) Iteratively Drilling Down – Use for loops to iterate over every child of a node, regardless of what the tag is. Repeat with additional nested for loop to drill further down the hierarchy.

(2) Drill down, one level at a time, through successive rounds of .find()/.findall() to select specific nodes which match a pattern. By default, the patterns only search immediate children.

(3) Use .find()/.findall() but provide an xpath search pattern to navigate straight to the desired node.

Searching a Tree: .find() and .findall()

- **.find()** - Finds the first child matching by tag name or path.
- **.findall()** - Finds all matching children, Returns as list.
- **.find()/.findall()** will only search immediate children unless an xpath is used: https://docs.python.org/3/library/xml.etree.elementtree.html#supported-xpath-syntax
- If no matches can be found, then .find() will return *None* type, and .findall() will return an *empty list*.

Accessing Attributes

```
<?xml version = "1.0" encoding = "UTF-8"?>
<snacks>
    <cookies>
        <cookie sellby="03/14/2020">chocolate chip</cookie>
        <cookie sellby="03/13/2020">oatmeal raisin</cookie>
        <cookie sellby="03/15/2020">peanut butter</cookie>
    </cookies>
</snacks>
```

```
import xml.etree.ElementTree as ET
tree = ET.parse("demo1.xml")
root = tree.getroot()

list_of_cookies = root.findall(".//cookie")
for cookie in list_of_cookies:
    print(cookie.text)
    print(cookie.get("sellby"))
```

Output:

```
chocolate chip
03/14/2020
oatmeal raisin
03/13/2020
peanut butter
03/15/2020

Process finished with exit code 0
```

Searching Attributes

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<snacks>
    <cookies>
        <cookie sellby="03/14/2020">chocolate chip</cookie>
        <cookie sellby="03/13/2020">oatmeal raisin</cookie>
        <cookie sellby="03/15/2020">peanut butter</cookie>
    </cookies>
</snacks>
```

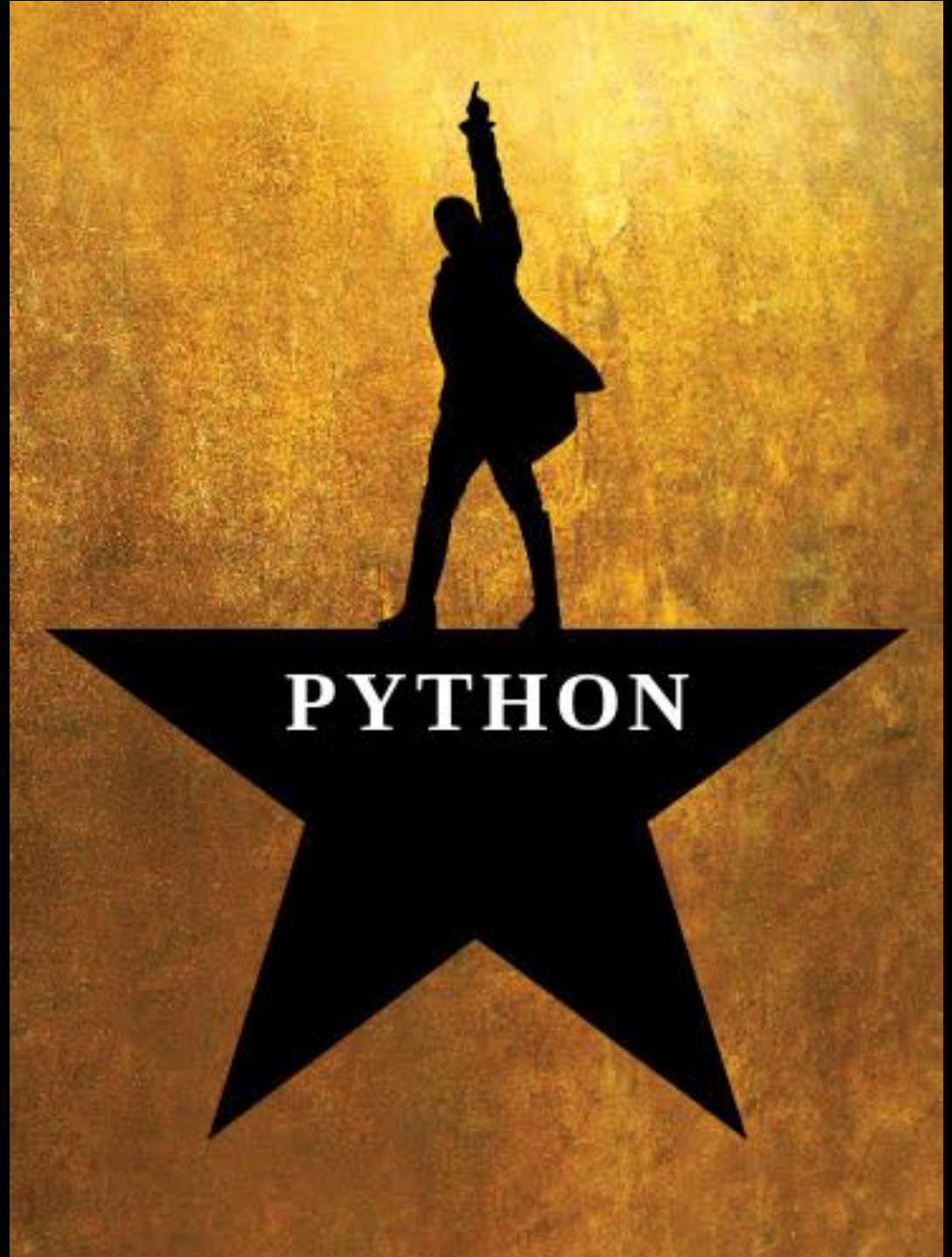
```
import xml.etree.ElementTree as ET
tree = ET.parse("demo1.xml")
root = tree.getroot()
# notice how the inner value of the attribute in the search is inside single quotes
this_cookie = root.find(".//cookie[@sellby='03/15/2020']")
print("Cookie batch with sell by date of 03/15/2020: " + this_cookie.text)
```

Output:

```
Cookie batch with sell by date of 03/15/2020: peanut butter

Process finished with exit code 0
```

Case 2: Fetching Data Over an API

(application programming interface)



1

API – Functionality for letting two kinds of software talk to each other.

2

Web Services – An implementation of an API for transferring data over the web

3

You will often hear these terms used **interchangeably**.

Dealing with Jargon

API Examples

- NASA Open APIs:
  https://api.nasa.gov/

- PubMed:
  https://www.ncbi.nlm.nih.gov/home/develop/api/

- Elsevier Scopus / Science Direct/ SciVal (Scholarly literature and journal articles):
  https://dev.elsevier.com/

- Library of Congress:
  https://labs.loc.gov/lc-for-robots/

- Twitter:
  https://developer.twitter.com/en/doc

More Jargon

- An API **request** will start with a **base url** (hostname + api path)
- An API can have multiple **endpoints** -- the locations from which you request different resources.
- Request is fleshed out with **parameters** that tell the server the specific information you are asking for.
- A **rate limit** or **quota** refers to how many requests you are allowed to make in a certain amount of time.
- Some services require you to aquire an **API Key** (secret passcode) to use the service.

Anatomy of a Web Services Request: Elsevier

https://api.elsevier.com/content/search/scopus?apiKey=my_api_key&query=AF-ID(60024266) AND PUBYEAR = 2018

Base url: https://api.elsevier.com/content/search

Endpoint: /scopus

Parameters:

apiKey=my_api_key

query=AF-ID(60024266) AND PUBYEAR = 2018

Anatomy of a Web Services Request: Elsevier

https://api.elsevier.com/content/search/scopus?apiKey=my_api_key&query=AF-ID(60024266) AND PUBYEAR = 2018

? separates the base url and endpoint from query parameters.

& separates multiple parameters from each other. It designates the end of one parameter and the start of another.

Anatomy of a Web Services Request: Pubmed OAH-PMH

https://www.ncbi.nlm.nih.gov/pmc/oai/oai.cgi?verb=ListRecords&from=2019-02-01
&until=2019-10-01&set=bmcbioc&metadataPrefix=oai_dc

Base url: https://www.ncbi.nlm.nih.gov/pmc *(API I want to use)*

Endpoint: /oai/oai.cgi *(The resource communication point I'm interested in)*

Parameters:

verb=ListRecords *(From OAI-PMH standard, what I'm asking for)*

from=2019-02-01 *(Start date for interval of records I want)*

until=2019-10-01 *(End date for the records I'm interested in)*

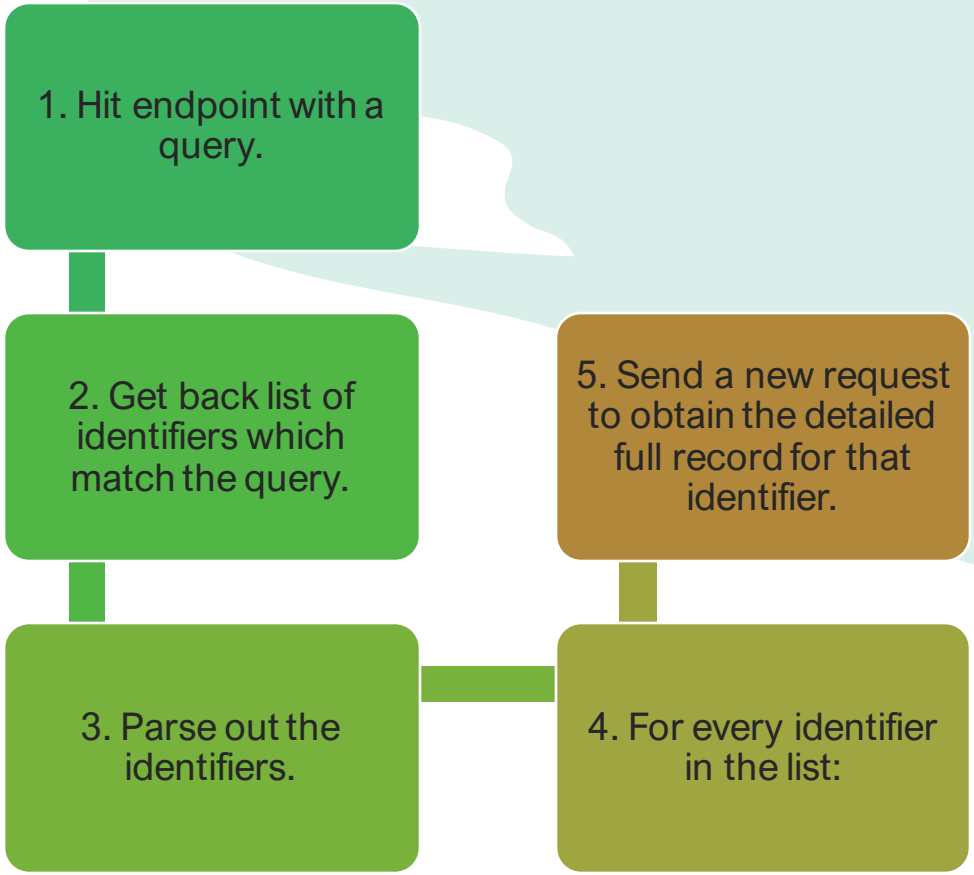
set=bmcbioc *(Only records in this category)*

metadataPrefix=oai_dc *(The metadata format I want results in)*

Some Answers to find in an API's Documentation

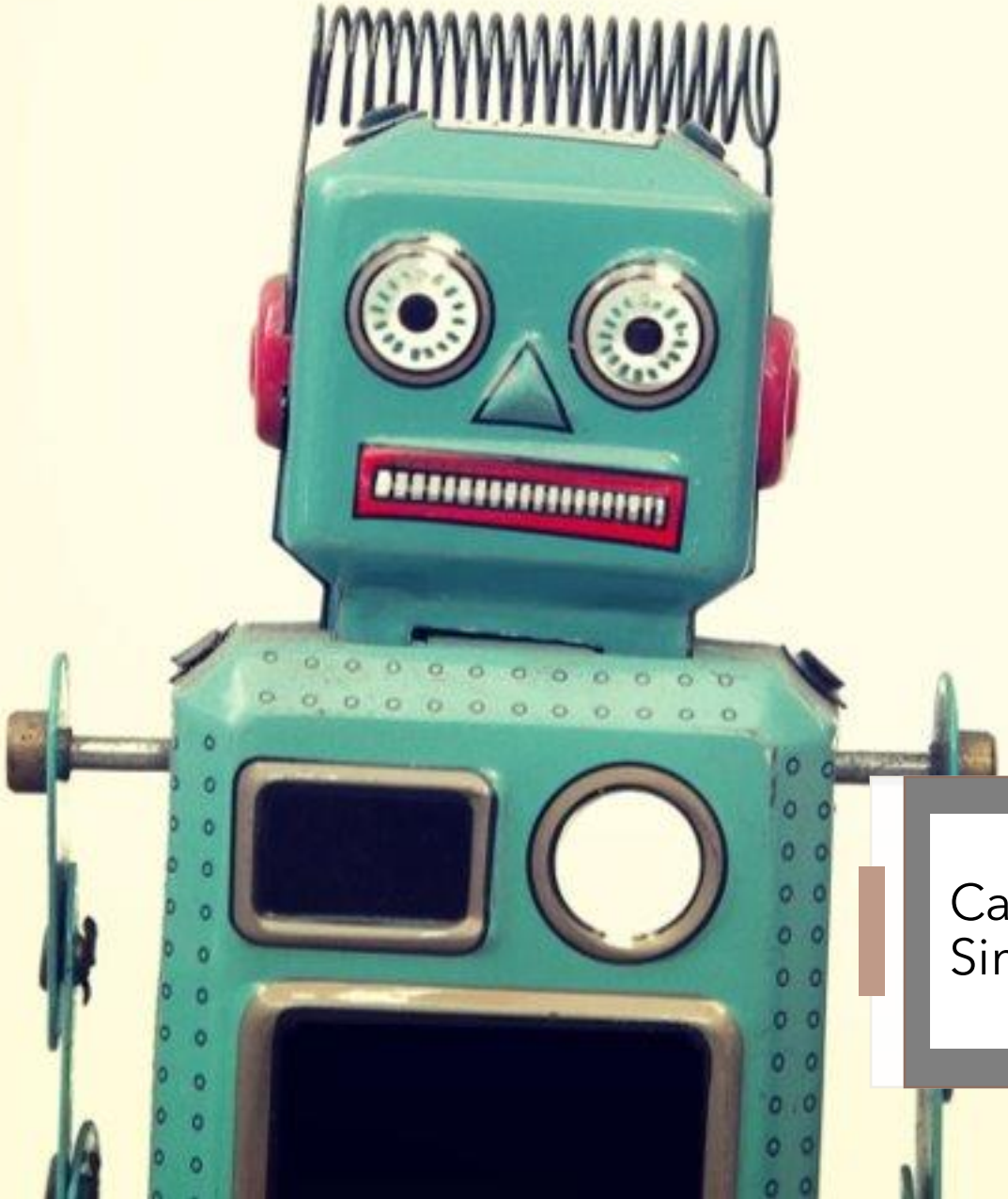
- Is there authentication and how does it work?
  - API Key?
  - IP-restricted?
- What are the major end points and what are their parameters?
- Is there a rate limit / quota?
- How does pagination work?
  - How many records per request.
  - How to cycle through all records that match your query when the number of results exceeds the number that you can fetch at one time.

A Common Workflow for APIs



case-2.py

- Use NASA public API to capture data about near earth objects.



Case 3
Simple Web Scraping

Requests
+ Beautiful Soup

Web Scraping

- Web scraping usually refers to extracting data from web pages, often in a fashion that simulates human behavior.

- Web scraping often entails heavy reliance on string-processing functions to clean data extracted from HTML, which is far more unruly and wild than the neatly structured machine-readable content exposed in APIs.

About Beautiful Soup 4

- To parse a document, pass either a string or a file handler to the BeautifulSoup constructor
- Content is first converted to Unicode.
- Transforms a complex HTML document into a complex tree of Python objects.
- The 4 objects you will interact with are:
  - **Tag** – Corresponds to an html or xml element; Has name and attributes
  - **NavigableString** – Contains text within a tag
  - **BeautifulSoup** – represents the parsed document as a whole
  - **Comment** – special type of NavigableString

https://www.crummy.com/software/BeautifulSoup/bs4/doc/

bs4 parser options

This table summarizes the advantages and disadvantages of each parser library:

Parser	Typical usage	Advantages	Disadvantages
Python's html.parser	`BeautifulSoup(markup, "html.parser")`	• Batteries included • Decent speed • Lenient (As of Python 2.7.3 and 3.2.)	• Not as fast as lxml, less lenient than html5lib.
lxml's HTML parser	`BeautifulSoup(markup, "lxml")`	• Very fast • Lenient	• External C dependency
lxml's XML parser	`BeautifulSoup(markup, "lxml-xml")` `BeautifulSoup(markup, "xml")`	• Very fast • The only currently supported XML parser	• External C dependency
html5lib	`BeautifulSoup(markup, "html5lib")`	• Extremely lenient • Parses pages the same way a web browser does • Creates valid HTML5	• Very slow • External Python dependency

https://www.crummy.com/software/BeautifulSoup/bs4/doc/

BS4: Navigating the Tree

- Tags may contain strings and other tags as children.
- Can navigate using tag names (such as `soup.p` or `soup.title`)
- Can access tag attributes similarly to a dictionary. (`tag['id']`)
- `soup.find_all('p')` <- Would return a set of tags which match p (paragraph)
- All a tags childrens are available in a list as a property of the tag called .contents
- Can also access children of a tag through a generator:
  `for child in title_tag.children:`

https://www.crummy.com/software/BeautifulSoup/bs4/doc/

Parse Elements

```
for child in soup.children:
    print(child)
```

```
<!DOCTYPE html>
<html lang="en">
<head>…  </head>
<body>
   <h2>Top</h2>
   <p class="top">This is the first paragraph</p>
   <p class="top">This is the second
paragraph</p>
   <h2>Middle</h2>
   <p>This is the <b>middle</b> paragraph</p>
   <h2>Bottom</h2>
   <p class="bottom">This is the fourth
paragraph</p>
   <p class="bottom">This is the fifth
paragraph</p>
</body>
</html>
```

Parse Elements

```
paragraphs = soup.find_all('p')
for p in paragraphs:
    print(p.text)
    if p.b:
        print("{} is in BOLD!".format(p.b))
```

```
<!DOCTYPE html>
<html lang="en">
<head>...  </head>
<body>
   <h2>Top</h2>
   <p class="top">This is the first paragraph</p>
   <p class="top">This is the second
paragraph</p>
   <h2>Middle</h2>
   <p>This is the <b>middle</b> paragraph</p>
   <h2>Bottom</h2>
   <p class="bottom">This is the fourth
paragraph</p>
   <p class="bottom">This is the fifth
paragraph</p>
</body>
</html>
```

case-3.py

- Introduction to web-scraping a sample file: http://volweb.utk.edu/~mihli1/demos/case3.html
- Obtain web page using requests.
- Scrape sample content with beautiful soup and HTML parser.

Web Scraping with Selenium

- Selenium is a package for automated testing of web pages. It replicates user behavior like clicking, filling out form data, pausing, etc.
- It has also become quite popular for web scraping due to reasons such as ability to interact with pages dynamically, allow JavaScript to execute.

Capturing Raw Data Over HTTP

REQUESTS

- Faster capture over HTTP.
- Simpler to implement.
- Can't interact with page.
- Won't run javascript.
- No default user-agent.
- Best choice for web services.

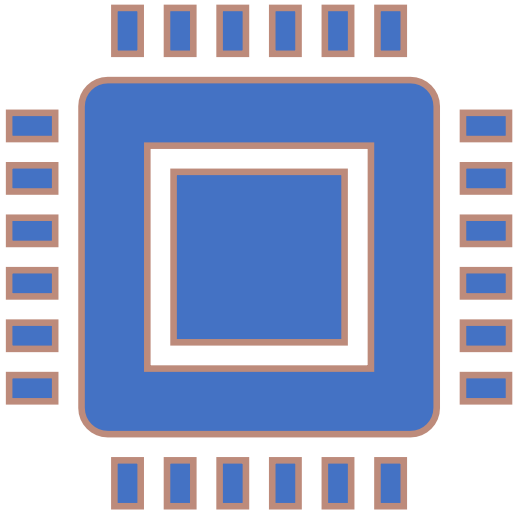
SELENIUM/WEB DRIVER

- Can interact with webpage to get to desired data such by navigating and clicking buttons.
- Will execute JavaScript.
- Slower than requests because you are operating through a browser.
- Browser's default user-agent
- Best choice for web pages with JS or needing interaction.

Web Scraping w/ Selenium

- Sometimes web pages aren't fully rendered in the moment you make the request. A page with JavaScript can take time to finish processing and loading the content.
- This means that whatever you get back from that HTTP request isn't going to have the full page content.
- Solution: Use Selenium & chromedriver-- Take control of a browser (Chrome) programmatically using a special driver.
- Allows page to fully load before retrieving its contents.

Web Scraping: The Complete Package



- **Selenium** – Python package implements tools for controlling a web driver.
- **Chromedriver** (or other web driver) - tool for automated testing of webapps. It supports navigating to web pages, user input, executes JavaScript, etc.
- **Google Chrome** – It is possible with other browsers but this is what we will use.
- **Beautiful Soup** – a Python library for pulling data out of HTML and *XML files.

** Note: If you want to use for XML parsing, will need to install lxml package*

Create Web Driver & Fetch Page Source

```
url = "http://kitty.ninja/demo.html"
driver = webdriver.Chrome(executable_path="~/Downloads/webdrivers/chromedriver")
driver.get(url)
```

Store Document Tree in Memory

```
soup = BeautifulSoup(driver.page_source, "html.parser")
```

case-4.py

- Example of scraping web page that uses javascript: http://volweb.utk.edu/~mihli1/demos/case4.html
- Launch chromedriver using selenium to control browser and interact with page.
- Load the source code of the browser being controlled into beautiful soup and proceed to parse as usual.